

NAYA, JAMES MBABILA

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EDUCATION

University of Massachusetts Boston

PhD, Integrative Biosciences, Biophysics Track

USA

Present

University of Ghana

Bachelor of Science in Engineering, Biomedical Engineering | First Class Honours

GPA 3.82 / 4.00

Capstone Project: Design of Novel Aspirator Connector for Fluid Evacuation during Cesarean Sections

Relevant Coursework: Medical Imaging, Medical Signal & Image Processing, Biophotonics, Bioinstrumentation, Biomedical Engineering Systems, Telemetry and Telemedicine, Medical Physics, Bioinformatics, Tissue Engineering

Honors & Leadership: Dean's Honor List; OBED Fellow; President, GSBE-UG; Speaker of Parliament; Electoral Commission Chair, ESUG

Legon, Ghana

Sep 2025

Nandom Senior High School

General Science Program

Highlights: Principal Contestant, National Science & Maths Quiz (NSMQ 2021); Contestant, Senior High School Renewable Energy Challenge (SHSREC 2021); Health Prefect (2020/2021)

UWR, Ghana

2018 – 2021

EXPERIENCE

Institute for Innovation and Next-Generation Engineering (INNOGEN)

Accra, Ghana

Founder & Operations Manager

Oct 2025 – Present

- Founded INNOGEN to deliver technical training, research opportunities, and innovation initiatives in biomedical engineering and accessible healthcare technology in Ghana.
- Spearheaded the inaugural multi-day virtual workshop "Building Low-Cost X-Ray Machines" (Dec 2025), guiding participants through medical imaging physics, component design, open-source modeling, and local fabrication strategies to reduce costs and enable Ghana's entry into the medical device value chain.

Department of Biomedical Engineering, University of Ghana

Accra, Ghana

Teaching Assistant (Biophysics, Biophotonics, Bioelectronics, Medical Physics)

Jan 2025 – Aug 2026

- Delivered tutorials and conducted laboratory sessions across assigned courses
- Taught 8 classes of the semester-long course BMEN 423 (Biophysics) to Level 400 students, as part of Prof. Elvis Tiburu's course
- Led Biophysics students on 3 academic tours of NMR, Flow Cytometry, and Mass Spectroscopy labs at the University of Ghana.
- Assisted lecturers with grading, assessment and preparing for classes.

School of Engineering Sciences, University of Ghana

Accra, Ghana

Editorial Board Member

Nov 2025 – Aug 2026

- Served as teaching and executive assistant to Dean Prof. Elsie Effah Kaufmann
- Curated Engineering Times 2026, the School's official magazine, alongside numerous newsletters and communications from the School.

- Helped organize and moderate major school events, including World Engineering Day (Moderator, panel discussion).
- Managed the School's social media handles, including X, Instagram, and LinkedIn, achieving significant growth across all platforms; LinkedIn followers grew by more than 300% (2,500 to 7,500) over the tenure.

Echo Change

Kuala Lumpur, Malaysia

Fully-Funded Global Delegate, Future Action Summit Malaysia 2025 | Nov 27 – 30, 2025

- Selected as one of the top ten fully-funded participants from a global pool of thousands of applicants.
- Led working group on Medical Technology and Policy, contributing to the development of the Post-2030 Youth Charter for Global Development.
- Engaged in discussions on Sustainable Development Goals (SDGs) with international changemakers and leaders.
- Participated in workshops, simulations, and cultural exchanges to advance youth-driven solutions for sustainability.

OBED Fellowship

Korle Bu Teaching Hospital, Accra

2024 Fellow, Global Health Design Initiative, University of Michigan & University of Ghana *Aug 2024 – Sep 2024*

- Conducted a comprehensive needs assessment within the Obstetrics and Gynecology Department through ethnographic observation of surgeries and clinician ward rounds, complemented by interviews and stakeholder engagements to identify medical and engineering challenges.
- Assessed 60 unique problems, developed design specifications, and selected one for a capstone project.
- Supervised by Ing. Prof. Elsie Effah Kaufmann, Dr. Claude Fiifi Hayford, Dr. Dhanalakshmi Thiagarajan (University of Michigan), and Prof. Julia Kramer (University of Michigan).

Komfo Anokye Teaching Hospital

Kumasi, Ghana

Biomedical Engineering Intern

May 2023 – Aug 2023 & Feb 2024 – Apr 2024

- Participated in calibrating, troubleshooting, servicing, repairing, and maintaining medical equipment including X-ray, MRI, ventilators, dialysis and anesthesia machines.
- Programmed and deployed medical equipment inventory software (EquiBase) for the Biomedical Engineering Department to assist in device inventory management.

Powerhouse Hospital

Kumasi, Ghana

Medical Laboratory Intern

Oct 2022 – Dec 2022

- Performed laboratory analyses on biological samples (hematology, clinical chemistry, microbiology, serology) and conducted procedures such as blood cross-matching and microscopic examinations using diagnostic equipment to support patient diagnoses.

RESEARCH

- Computer Vision-Based Health Status Assessment and Transmission of Ocular Images: The Na- β iPlatform (Course Project, BMEN 416, Telemetry and Telemedicine, *Jun 2025 – Oct 2025*)
- Design of Novel Aspirator Connector for Fluid Evacuation during Cesarean Sections (Course Project, BMEN 400, Capstone Design, *Nov 2024 – Aug 2025*)
- Using Generative AI to Study the Molecular Structure of Plant Metabolites Present in Locally Produced Herbal Medicines Used for Treating Malaria (Mini-Ph.D. Program, Reachsci, USA, *Nov 2023 – Jul 2024*)
- Evaluating Antimicrobial and Pharmacokinetic Effects of Using Cyanogenic Glycosides from Cassava for Potential Applications in Human-Machine Interface (HMI) Infections (Course Project, BMEN 315, Research Methods, *Oct 2022 – Dec 2023*)
- Design of a Biosand Filter for Water Purification and Sustainability (Course Project, BMEN 306, Design and Selection of Biomaterials, *May 2024 – Aug 2024*)
- Design of an Elbow Splint for Patients with Cerebral Palsy (Course Project, BMEN 308, Design of Mechanical Systems, *May 2024 – Aug 2024*)
- Design of an Echolocation Glasses for the Blind (Course Project, BMEN 202, Introduction to Biomedical Engineering, *March 2023 – Jun 2023*)

CONFERENCES & PRESENTATIONS

- **ICAST 2025 – 20th International Student Conference on Advanced Science and Technology Satellite** (Satellite/Online Session) | Co-organized by Kumamoto University (Japan) and National Chiayi University (Taiwan)
Oral Presentation | Dec 11, 2025 | *"Design of Novel Medical Aspirator Connector for Cesarean Sections: Enhancing Efficiencies and Reducing Surgical Risks"* | **Awarded Certificate of Presentation**
- **Medical Innovation Expo 2025** | Academic City University, Accra, Ghana
Poster Presentation | Oct 2025 | *"Computer Vision-Based Health Status Assessment and Transmission of Ocular Images: The Na- β iPlatform"* | **Awarded Certificate of Presentation**
- **Advancing Computational Neuroscience Through Collaborative Learning** | *Global Webinar Series*
Moderator | Apr 2025
 - Moderated Days 4 & 5 of a 5-day international webinar series featuring speakers from University College London, University of Ghana, and Yale School of Medicine.
 - Supported by Allen Institute, Ghana Society of Biomedical Engineers, Ghana Neuroscience Society, and University of Amsterdam.
- **4th National Ghana Society of Biomedical Engineers Conference** | Accra, Ghana
Poster Presentation | Oct 2024
"Design of an Echolocation Glasses for the Blind." | **Best Project Presentation.**

LEADERSHIP & ACTIVITIES

Engineering Students Association of University of Ghana (ESUG)

Legon, Accra

Nov 2024 – Sep 2025

• Speaker of Parliament

- Presided over parliamentary sessions for the 19th ESUG Administration during General Assemblies to facilitate productive legislative discourse and adherence to constitutional procedures.
- Facilitated policy approvals and resolutions for 1,000+ engineering students.

Sep 2023 – Aug 2024

• Electoral Commission Chair

- Conducted free, fair, and incident-free elections to select new executive members to replace the 18th ESUG Administration

Jun 2023 – Aug 2023

• 2nd Deputy Electoral Commissioner

- Assisted in conducting free and fair elections to select new executive members to replace the 17th ESUG Administration

Street Fund Global Foundation

Ghana

Jan 2025 – Aug 2026

Head, Project Department

- Organized and led an educational presentation to the Madina La Nkwantanang School B on career pathways and academic performance improvement
- Led planning and implementation of community-focused social impact initiatives, including Streetfund's annual Streetmas
- Organized project management seminar for the foundation to learn from field experts

Ghana Society of Biomedical Engineers – University of Ghana

Legon, Accra

Aug 2023 – Aug 2024

President, GSBE- UG

- Spearheaded the Ghana Society of Biomedical Engineers – University of Ghana (GSBE-UG) as its 2nd President and enhanced member engagement and organizational reach over a 1-year tenure.
- Designed and implemented diverse programs, including a peer mentorship initiative, the Learning Space Initiative, 3 industrial field trips, and multiple professional development seminars
- Pioneered the maiden Nationwide Biomedical Engineering Webinar in collaboration with 7 universities, bringing together over 700 students to learn from each other and from experts through a seminar series.

Nandom Senior High School

Nandom, UWR, Ghana

National Science and Maths Team

2021

- Served as principal contestant for the school and led the team through both internal and regional contests

Science and Maths Club Overseer

2019 – 2021

- Coordinated activities of the club including organizing interclass competitions, and recruiting new members for the National Science and Maths Team.

Health Prefect

2020 – 2021

- Led health policies and coordinated student welfare initiatives.

Multicultural Group Roles

2018 – 2020

- Vice President of the Southern Students Union
- Financial Secretary of the Upper East Students Union

AWARDS, FELLOWSHIPS & RECOGNITIONS

- 2nd Best Student, Department of Biomedical Engineering Class of 2025
- Head of Department Recognition Award for Distinguished Leadership (Jul 2025)
- Best Project Presentation Award, 4th National GSBE Conference (Oct 2024) | Topic: *"Design of Echolocation Glasses for Visually Impaired Individuals"*
- Dean's Honor List (2023/2024 Academic Year), School of Engineering Sciences
- Educational Pathways International Scholarship (Jan 2022, Jan 2023, Oct 2023, Oct 2024) | approximately GHC 40,000.00 (\$3,600.00 equivalent)
- Overall Best Student, Nandom Senior High School (2018/19 and 2019/20 Academic Years)

CERTIFICATES

- Google Advanced Data Analytics Certificate Badge (Coursera & Google, 2025)
- Google Project Management Professional Certificate Badge (Coursera & Google, 2024)
- Hospital Planning & Design and Equipment Management (Hospital Engineering Academy, November 2024)
- Particle Physics: An Introduction (Coursera & Université de Genève, Nov 2023)
- Quantum Computing (Qubit by Qubit, The Coding School & IBM, Oct 2022 – Apr 2023)
- Virtual Immersion in Biomedical Engineering (Worcester Polytechnic Institute, Summer Program 2022)

Additional certifications (39 total) available upon request or via LinkedIn/portfolio.

SKILLS

Technical

- Engineering & Design
- CAD (SolidWorks, Autodesk Fusion)
- Medical Equipment Maintenance & Calibration
- Engineering Design & Research
- Laboratory Equipment Operation

Programming & Technology

- Python (Web Development, Computer Vision, Data Science & Machine Learning)
- MATLAB
- Arduino & Proteus Design Suite
- Web & Application Development
- SIEM & Cybersecurity Fundamentals

MEMBERSHIP IN PROFESSIONAL ASSOCIATIONS

- Ghana Institution of Engineering | Student Member (2024-Present)
- The Field Engineer | Student Member (2023-Present)
- Elsie Effah Kaufmann Foundation | Volunteer (2023-Present)
- Vispad Institute of Technology, Ghana | Fellow and Teaching Assistant, (2022-Present)

MEDIA APPEARANCES & FEATURES

- NSMQ Superstar Spotlight by Prof. Elsie Effah Kaufmann for excellence. → [LinkedIn Post](#)
- Life Studying Biomedical Engineering in Africa → [The Field Engineer Article](#)
- GSBE-UG Healthcare Revolution → [Elevate HTM Post](#)
- Hybrid Power Plant Presentation at the SHS Renewable Energy Challenge → [Home Radio Post](#)
- Quantum Computing Capstone Project Showcase → [YouTube Video](#)

INTERESTS & LANGUAGES

Interests: STEM documentaries, Leadership, History, Poetry, Reading, Mentorship

Languages: English (Official Language), Local Languages: Nabdam (fluent), Twi (fluent)